



LUND
UNIVERSITY

Numerical Analysis for E3
Numerical Analysis, Centre for Mathematical Sciences

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Homework assignment 6

Delivery date: Tuesday, May 20th., not later than 18:00.

Topics of this assignment: Newton's method; order of convergence; initial value problems; Euler explicit and implicit methods.

1. Turbo Newton. In this exercise we will consider a higher order iterative method to solve numerically the problem $f(x) = 0$. Newton's method is based on a first order approximation of the function $f(x)$. To obtain a higher order method we instead substitute $f(x)$ by a second order approximation and compute the roots of this parabola

$$f(x) \approx f(x_n) + f'(x_n)(x - x_n) + \frac{f''(x_n)}{2}(x - x_n)^2 = 0.$$

This problem has two possible solutions

$$\Delta x_{\pm} = (x - x_n)_{\pm} = \frac{-f'(x_n) \pm \sqrt{f'(x_n)^2 - 2f''(x_n)f(x_n)}}{f''(x_n)}.$$

We choose the root that lies nearer to x_n . A detailed case study shows that the Turbo-Newton increment is given by

$$\Delta x_n = \frac{-f'(x_n) + \text{sign } f'(x_n) \sqrt{f'(x_n)^2 - 2f''(x_n)f(x_n)}}{f''(x_n)}.$$

The iteration is defined as usual

$$x_{n+1} = x_n + \Delta x_n.$$

There are two details one has to take care of when implementing the method:

- The increment Δx_n has to be real and therefore it is a requirement that $f'(x_n)^2 > 2f''(x_n)f(x_n)$.
- The turbo-Newton increment is not defined for $f''(x_n) = 0$.

When one of this situations occurs, we just is to use the classical Newton increment

$$\Delta x_n = -\frac{f(x_n)}{f'(x_n)}$$

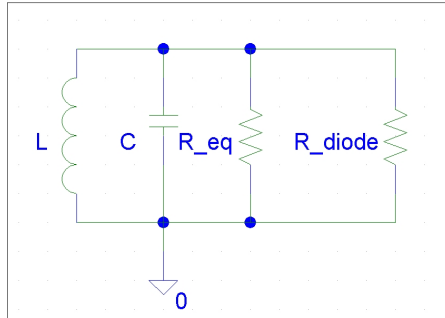
and continue the iterations.

Task 1a): Implement the method in a file `turbonewton.m` that has as parameters f, f', f'' and a tolerance tol , and returns the numerical approximation x to the solution of $f(x) = 0$.

Hint: Use the "skeleton" of the Exercise 1 in Homework 5 as starting point.

Task 1b): Approximate the value of $\log \pi$ up to full precision. This can be done searching the root of the function $f(x) = e^x - \pi$ and using $x_0 = 1.096629$. Compare with the reference value 1.14472988584940016 and study the convergence of the method. Is it faster than the classical Newton method? Motivate your answer.

2. Tunnel Diode Oscillator. Consider the following tank circuit



The components are an inductance L , a capacitance C , the resistance denominated R_{eq} representing the total parasitic resistance of the induction and the capacitor together, and finally the resistance R_{diode} which is the AC model resistance of a diode. For common types of diodes this resistance is positive and this contributes even more to the damping of the tank system. But the situation is different when the diode is a *tunnel diode*, because this type can have negative resistance.

The differential equation that governs the voltage of the circuit with a tunnel diode is

$$\ddot{V}(t) - \alpha \dot{V}(t) + 3\beta V^2(t) \dot{V}(t) + \frac{1}{LC} V(t) = 0$$

with parameters α, β which depend on the diode and the parasitic resistance R_{eq} .

Task 2a): Reduce the differential equation to a first system of the form $\dot{Y}(t) = F(t, Y)$.

Task 2b): Solve the differential equation using the implicit Euler method as corrector and explicit Euler method as predictor with just 2 fixed point iterations. Make two simulations in the time interval $[0, 0.0003]$, one with the set of values $\alpha = 1e6$, $\beta = 10^6$, $L = 10^{-5}$ and $C = 10^{-5}$ and the other with $\alpha = 10^4$, $\beta = 10^6$, $L = 10^{-5}$ and $C = 10^{-5}$. Use as initial conditions $\dot{V}(0) = 0$ and $V(0) = 0.2$ and step size $h \leq 10^{-7}$. Observe the differences in behavior and make conclusions.

Task 2c): The difference between the explicit and the implicit Euler method is an estimate for the local error at each step. Plot this difference (use the command `errorbar` as help for the visualisation). Comment your observations.